

● MEDIUM

● ALGEBRA

● ANSWER KEY

Linear functions

03

10 answers · Rationale-rich · Teach from doc

ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1**3500**

The correct answer is 3500. The given information includes a cost, \$800, to produce 100 shirts. Substituting $c(x) = 800$ and $x = 100$ into the given equation yields $800 = m \cdot 100 + 500$. Subtracting 500 from both sides of the equation yields $300 = m \cdot 100$. Dividing both sides of this equation by 100 yields $3 = m$. Substituting the value of m into the given equation yields $c(x) = 3x + 500$. Substituting 1000 for x in this equation and solving for $c(x)$ gives the cost of 1000 shirts: $3(1000) + 500$, or 3500.

Q2

C

C. $y = 8x + 2$

Choice C is correct. It's given that the relationship between x and y is linear. An equation representing a linear relationship can be written in the form $y = mx + b$, where m is the slope and b is the y -coordinate of the y -intercept of the graph of the relationship in the xy -plane. It's given that for every increase in the value of x by 1, the value of y increases by 8. The slope of a line can be expressed as the change in y over the change in x . Thus, the slope, m , of the line representing this relationship can be expressed as $\frac{8}{1}$, or 8. Substituting 8 for m in the equation $y = mx + b$ yields $y = 8x + b$. It's also given that when the value of x is 2, the value of y is 18. Substituting 2 for x and 18 for y in the equation $y = 8x + b$ yields $18 = 8(2) + b$, or $18 = 16 + b$. Subtracting 16 from each side of this equation yields $2 = b$. Substituting 2 for b in the equation $y = 8x + b$ yields $y = 8x + 2$. Therefore, the equation $y = 8x + 2$ represents this relationship.

Choice A is incorrect. This equation represents a relationship where for every increase in the value of x by 1, the value of y increases by 2, not 8, and when the value of x is 2, the value of y is 22, not 18.

Choice B is incorrect. This equation represents a relationship where for every increase in the value of x by 1, the value of y increases by 2, not 8, and when the value of x is 2, the value of y is 12, not 18.

Choice D is incorrect. This equation represents a relationship where for every increase in the value of x by 1, the value of y increases by 3, not 8, and when the value of x is 2, the value of y is 32, not 18.

Q3**A****A.** 21

Choice A is correct. The x -intercept of a graph in the xy -plane is the point on the graph where $y = 0$. It's given that function h is defined by $hx = 4x + 28$. Therefore, the equation representing the graph of $y = hx$ is $y = 4x + 28$. Substituting 0 for y in the equation $y = 4x + 28$ yields $0 = 4x + 28$. Subtracting 28 from both sides of this equation yields $-28 = 4x$. Dividing both sides of this equation by 4 yields $-7 = x$. Therefore, the x -intercept of the graph of $y = hx$ in the xy -plane is $-7, 0$. It's given that the x -intercept of the graph of $y = hx$ is $a, 0$. Therefore, $a = -7$. The y -intercept of a graph in the xy -plane is the point on the graph where $x = 0$. Substituting 0 for x in the equation $y = 4x + 28$ yields $y = 4(0) + 28$, or $y = 28$. Therefore, the y -intercept of the graph of $y = hx$ in the xy -plane is $0, 28$. It's given that the y -intercept of the graph of $y = hx$ is $0, b$. Therefore, $b = 28$. If $a = -7$ and $b = 28$, then the value of $a + b$ is $-7 + 28$, or 21.

Choice B is incorrect. This is the value of b , not $a + b$.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of $-a + b$, not $a + b$.

Q4**D****D.** 4

Choice D is correct. It's given that the function w models the volume of liquid, in milliliters, in a container t seconds after it begins draining from a hole at the bottom. The given function $w(t) = 300 - 4t$ can be rewritten as $w(t) = -4t + 300$. Thus, for each increase of t by 1, the value of $w(t)$ decreases by $4(1)$, or 4. Therefore, the predicted volume, in milliliters, draining from the container each second is 4 milliliters.

Choice A is incorrect. This is the amount of liquid, in milliliters, in the container before the liquid begins draining.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Q5**A**

A. Each game costs \$25.

Choice A is correct. The given graph is a line, and the slope of a line is defined as the change in the value of y for each increase in the value of x by 1. It's given that y represents the total cost, in dollars, and that x represents the number of games. Therefore, the change in the value of y for each increase in the value of x by 1 represents the change in total cost, in dollars, for each increase in the number of games by 1. In other words, the slope represents the cost, in dollars, per game. The graph shows that when the value of x increases from 0 to 1, the value of y increases from 100 to 125. It follows that the slope is 25, or the cost per game is \$ 25. Thus, the best interpretation of the slope of the graph is that each game costs \$ 25.

Choice B is incorrect. This is an interpretation of the y -intercept of the graph rather than the slope of the graph.

Choice C is incorrect. The slope of the graph is the cost per game, not the cost of the video game system.

Choice D is incorrect. Each game costs \$ 25, not \$ 100.

Q6**A**

A. The team will have moved all the cargo in about 4.8 hours.

Choice A is correct. The x -intercept of the line with equation $y = 120 - 25x$ can be found by substituting 0 for y and finding the value of x . When $y = 0$, $x = 4.8$, so the x -intercept is at $(4.8, 0)$. Since y represents the number of tons of cargo remaining to be moved x hours after the team started working, it follows that the x -intercept refers to the team having no cargo remaining to be moved after 4.8 hours. In other words, the team will have moved all of the cargo after about 4.8 hours.

Choice B is incorrect and may result from incorrectly interpreting the value 4.8. Choices C and D are incorrect and may result from misunderstanding the x -intercept. These statements are accurate but not directly relevant to the x -intercept.

Q7**D****D.** 44

Choice D is correct. It's given that $f(x) = x + 6$. Substituting 2 for x in the function f yields $f(2) = 2 + 6$, or $f(2) = 8$. It's also given that $g(x) = 6x$. Substituting 2 for x in the function g yields $g(2) = (6)(2)$, or $g(2) = 12$. Substituting 8 for $f(2)$ and 12 for $g(2)$ in the expression $7f(2) - g(2)$ yields $7(8) - 12$, or 44. Therefore, the value of $7f(2) - g(2)$ is 44.

Choice A is incorrect. This is the value of $f(2) - g(2)$, not $7f(2) - g(2)$.

Choice B is incorrect. This is the value of $f(2)$, not $7f(2) - g(2)$.

Choice C is incorrect and may result from conceptual or calculation errors.

Q8**B****B.** $hx = 2x + 7$

Choice B is correct. It's given that the graph of the linear function h , where $y = hx$, passes through the points 7, 21 and 9, 25 in the xy -plane. An equation defining h can be written in the form $y = mx + b$, where $y = hx$, m represents the slope of the graph in the xy -plane, and b represents the y -coordinate of the y -intercept of the graph. The slope can be found using any two points, x_1, y_1 and x_2, y_2 , and the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting 7, 21 and 9, 25 for x_1, y_1 and x_2, y_2 , respectively, in the slope formula yields $m = \frac{25 - 21}{9 - 7}$, which is equivalent to $m = \frac{4}{2}$, or $m = 2$. Substituting 2 for m and 7, 21 for x, y in the equation $y = mx + b$ yields $21 = 27 + b$, or $21 = 14 + b$. Subtracting 14 from each side of this equation yields $7 = b$. Substituting 2 for m and 7 for b in the equation $y = mx + b$ yields $y = 2x + 7$. Since $y = hx$, it follows that the equation that defines h is $hx = 2x + 7$.

Choice A is incorrect. For this function, the graph of $y = hx$ in the xy -plane would pass through 7, 0, not 7, 21, and 9, 1, not 9, 25.

Choice C is incorrect. For this function, the graph of $y = hx$ in the xy -plane would pass through 7, 70, not 7, 21, and 9, 84, not 9, 25.

Choice D is incorrect. For this function, the graph of $y = hx$ in the xy -plane would pass through 7, 88, not 7, 21, and 9, 106, not 9, 25.

Q9**C**

C. $fh = 65h + 90$

Choice C is correct. It's given that there is a linear relationship between a plumber's hours of labor, h , and the plumber's total charge fh , in dollars. It follows that the relationship can be represented by an equation of the form $fh = mh + b$, where m is the rate of change of the function f and b is a constant. The rate of change of f can be calculated by dividing the difference in two values of fh by the difference in the corresponding values of h . Based on the values given in the table, the rate of change of f is $\frac{285 - 155}{3 - 1}$, or 65. Substituting 65 for m in the equation $fh = mh + b$ yields $fh = 65h + b$. The value of b can be found by substituting any value of h and its corresponding value of fh for h and fh , respectively, in this equation. Substituting 1 for h and 155 for fh yields $155 = 65(1) + b$, or $155 = 65 + b$. Subtracting 65 from both sides of this equation yields $90 = b$. Substituting 90 for b in the equation $fh = 65h + b$ yields $f(h) = 65h + 90$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q10

C

C. $fx = 4x + 3$

Choice C is correct. In the xy -plane, an equation of the graph of a linear function can be written in the form $fx = mx + b$, where m represents the slope and b represents the y -intercept of the graph of $y = fx$. It's given that the graph of the linear function f , where $y = fx$, in the xy -plane contains the point $0, 3$. Thus, $b = 3$. The slope of the graph of a line containing any two points x_1, y_1 and x_2, y_2 can be found using the slope formula, $m = \frac{y_2 - y_1}{x_2 - x_1}$. Since it's given that the graph of the linear function f contains the points $0, 3$ and $7, 31$, it follows that the slope of the graph of the line containing these points is $m = \frac{31 - 3}{7 - 0}$, or $m = 4$. Substituting 4 for m and 3 for b in $fx = mx + b$ yields $fx = 4x + 3$.

Choice A is incorrect. This function represents a graph with a slope of 28 and a y -intercept of $0, 34$.

Choice B is incorrect. This function represents a graph with a slope of 3 and a y -intercept of $0, 38$.

Choice D is incorrect. This function represents a graph with a slope of 7 and a y -intercept of $0, 3$.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

Generated 2026-05-29 · cb-educator-question-bank